

Human thyroid organoids: tools for understanding thyroid diseases and development

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Abstract

The thyroid gland develops in five main stages, each characterized by specific molecular and cellular processes that are highly conserved across invertebrates and vertebrate species. In vivo studies have elucidated the key molecular players orchestrating each stage of thyroid organogenesis and thyroid hormone biosynthesis, laying the groundwork for developing thyroid organoids in vitro. Thyroid organoids have been successfully derived from mouse and human pluripotent stem cells, including embryonic stem cells and induced pluripotent stem cells, as well as from fetal and adult thyroid tissues. Protocols using directed differentiation through chemical signalling modulators and/or by overexpression of pivotal thyroid transcription factors have yielded thyroid progenitors with varying levels of maturation, differing in timeline, purity and functional outcomes. However, only a few protocols have succeeded in producing thyroid organoids capable of recapitulating thyroid functionality both in vitro and in vivo. This Review explores the various thyroid organoid models currently available, their contributions to understanding thyroid developmental biology and their application in modelling thyroid diseases. We also discuss the current challenges to translating thyroid organoid technology from experimental frameworks into clinical applications, framing the question: could thyroid organoids offer a therapeutic avenue for hypothyroidism, particularly in cases of congenital, post-surgical and treatment-refractory conditions?

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- Thyroid development involves intricately coordinated spatial and molecular mechanisms.
- Differentiation of pluripotent stem cells into functional thyroid organoids revisits every stage of tissue formation and offers a novel model for dissecting development and modelling diseases.
- Healthy and diseased human thyroid tissue-derived organoids are emerging as a rapid and efficient tool for addressing adult-onset conditions.
- The successful transplantation and functional rescue achieved with human thyroid organoids in mice highlight their therapeutic potential for a subset of patients with congenital, post-surgical or treatment-refractory hypothyroidism.
- Key translational challenges include achieving full in vitro maturation using transplantation-safe differentiation methods, ensuring immune compatibility and defining optimal transplantation sites.

Introduction

The thyroid gland synthesizes thyroid hormones essential for organ development and homeostasis¹. Within the past decade, advances in human thyroid organoids, derived from pluripotent and thyroid tissue, have provided in vitro models that recapitulate key developmental stages and thyroid physiology. This Review consolidates the current status of thyroid organoid technology, addressing its use in modelling diseases and exploring the potential of thyroid organoids for therapeutic transplantation. A central question remains: can thyroid organoids become viable functional grafts for restoring thyroid function in patients, especially those inadequately served by conventional hormone replacement?

Thyroid gland development: from specification to thyroid hormone synthesis

The thyroid gland derives from the anterior foregut endoderm (AFE) in vertebrates, and its development proceeds through six key stages: specification, budding from the endoderm layer and relocalization in front of the trachea, proliferation (also known as expansion), folliculogenesis and functional differentiation^{2–4} (Fig. 1). Each stage is characterized by specific molecular events and morphological changes, as discussed in the following subsections.

Endoderm specification

During gastrulation, coordinated activation and inhibition of WNT– β -catenin signalling, BMP and NODAL–TGF β signalling drive formation of the definitive endoderm^{5–10}. As gastrulation concludes, the definitive endoderm is patterned along the anterior–posterior axis to generate organs such as the thyroid, thymus, lung, liver, pancreas and gastrointestinal tract^{7,11}. Endodermal fate is further shaped by signals including FGFs, WNT and retinoic acid from neighbouring tissues^{12,13}. WNT– β -catenin activity is essential for midgut and hindgut specification, and its inhibition is required for foregut specification¹⁴. Notably, TGF β and BMP inhibition efficiently promote AFE induction from definitive endoderm-derived human pluripotent stem (PS) cells^{15–18}.

Thyroid specification

Thyroid development is first marked by thickening of the foregut endoderm epithelium to form the thyroid anlage, followed by cell specification into thyroid progenitors characterized by the co-expression of transcription factors NKX2.1, PAX8, FOXE1 and HHEX (at embryonic day 8.5 (E8.5) in mouse and 3 post-conception weeks (PCW) in humans)^{15–20} (Fig. 1). NKX2.1 regulates HHEX, FOXE1 and PAX8, and PAX8 and HHEX mutually regulate each other^{16,21,22}. A study conducted in 2023 in early mouse embryos revealed a sequential appearance of PAX8-positive cells at E7.5, followed by PAX8-positive, NKX2.1-positive thyroid progenitors at E8.5 (ref. 23). Although the molecular network is defined, the extrinsic signals guiding thyroid fate were unclear until studies using zebrafish and stem cell models highlighted the role of surrounding mesoderm, particularly cardiac mesoderm, as a source of FGF and BMP signals. Thyroid defects often coincide with cardiac malformations, underscoring this interaction^{23–32}. In vitro, PS cell-derived definitive endoderm treated with FGF2 and BMP4 can generate thyroid progenitors and mature follicles^{33–36}. Additionally, TAZ promotes PAX8 and NKX2.1 expression, supporting thyroid specification³⁷, whereas iodine excess could suppress their expression³⁸. In addition, thyroid morphogenesis depends on apical constriction driven by the proteins CDC42 and SHROOM3, and, without these regulators, the thyroid bud fails to form correctly, underscoring the essential role of cytoskeletal dynamics in early thyroid development³⁹.

Budding and relocalization

The thyroid primordium buds off the foregut endoderm (E9.5–E10 in mice, 4.5–6 PCW in humans), detaches and migrates to the final cervical position^{15,22,40} (Fig. 1). This recruitment-dependent growth of the thyroid bud is independent of the proliferation of NKX2.1-expressing progenitors³. The regulation of thyroid bud progression and migration is largely unknown, but studies suggest that SOX9 and FOXE1 expression and pseudostratification with cytoskeletal remodelling are required^{3,13,20,22,41–46}.

Proliferation (also known as expansion)

Initial proliferation of thyroid primordium cells occurs between E10.5 and E11.5 in mice and 6 and 7.5 PCW in humans, followed by bilobation, fusion with ultimobranchial bodies and, consequently, incorporation of C cell precursors^{3,46}. At this stage, proliferation becomes localized at the tip of the parenchymal cord, which will form the future follicle clusters, and gradually decreases as follicles mature^{33,46–48}. Unlike in adult thyroid follicular cells (TFCs), TSH does not regulate thyroid growth during development⁴⁷. In vitro, human PS cell-derived thyroid progenitors show high proliferation with cyclic AMP (cAMP) stimulation but reduced proliferation in response to TSH⁴⁸. Thyroid hypoplasia has been reported in SHH, TBX1 and FGF8-deficient mice, where it appears to arise primarily from mesoderm-derived defects affecting early thyroid expansion^{29,49,50}. A similar hypoplastic phenotype is observed in the absence of HES1 (ref. 30) or SOX9 and FGF10 expression⁴⁵, further highlighting the roles of these factors as intrinsic regulators of bud and thyroid growth and morphogenesis.

Folliculogenesis

The terminal differentiation step involves organization into follicular structures with apical–basal polarity (E13.5 in mice and 9 PCW in humans), essential for thyroid hormone synthesis and storage^{3,46,51,52}. Folliculogenesis depends on extrinsic signals from the extracellular matrix, mesenchymal cells, mesodermal cells and endothelium, with VEGFA,

laminins and BMP–SMAD signalling promoting lumen formation^{53–55}. Endothelial cells contribute to thyroid follicle formation by producing extracellular vesicles, which can be incorporated into TFC progenitors⁵⁴.

In vitro, primary thyrocytes or thyroid cell lines cultured in suspension form spherical aggregates historically described as follicular-like structures based on ultrastructural features; however, proper apical–basal polarity and fully functional follicular organization were not demonstrated. By contrast, more recent studies show that in 3D cultures within extracellular matrices (such as collagen or Matrigel), and particularly under TSH stimulation, thyrocytes reorganize into functional follicles^{34,48,56–58}. Apical–basal polarity is crucial: XB130 deficiency and PAX8 and/or CDH16 deficiency disrupt polarity and affect follicle formation^{59,60}, and MUC1 marks early lumen initiation⁴⁶. Thyroglobulin (TG) contributes to follicle enlargement but is not required for initial formation, indicating distinct mechanisms for morphogenesis and functional differentiation⁴⁶.

Differentiation and thyroid hormone synthesis

TG expression (E13.5) and folliculogenesis mark the onset of functional differentiation of the gland^{2–4,46}. Thyroid-stimulating hormone (TSH) receptor (TSHR) and thyroid peroxidase (TPO) start to be observed from E14.5 (refs. 2,61), followed by expression of sodium–iodide cotransporter (SLC5A5) and thyroid hormone biosynthesis at E16.5

(refs. 2,61). In humans, TG, TSHR and TPO are detected from 7 PCW⁴, whereas SLC5A5 is observed at 11 PCW, and tetraiodothyronine (T₄) production from 12 PCW⁴. The roles of thyroid transcription factors in regulating the expression of genes driving differentiation and function are well established^{21,62–65}. cAMP is the major intracellular second messenger of TSH action on thyroid cells, stimulating growth as well as the function and differentiation of thyrocytes^{47,66}. Thyroid hormone synthesis depends on proper follicular organization, iodide availability and expression of key thyroid hormone machinery genes. Iodide is absorbed from the gut, transported into thyrocytes via NIS^{67,68}, and effluxed to the lumen through pendrin, anoctamin1 (ANO1), and chloride voltage-gated channel 5 (CLC5)^{69–74}. In the apical membrane, TPO catalyses iodination of TG in an H₂O₂-dependent process^{75,76}, with DUOX2 as the main H₂O₂ source, assisted by DUOXA1 and DUOXA2 (refs. 77,78). Iodinated TG in the lumen forms tri-iodothyronine (T₃) and T₄ (refs. 79–81), which are released after endocytosis and proteolysis into thyrocytes^{82,83}, and iodide is recycled by DEHAL1 (ref. 84). Mechanisms involved in thyroid hormone synthesis have been extensively reviewed¹.

Thyroid organoids: translating development from in vivo to in vitro

Within the past two decades, animal models and 2D cell lines have been instrumental in studying organ development and disease mechanisms,

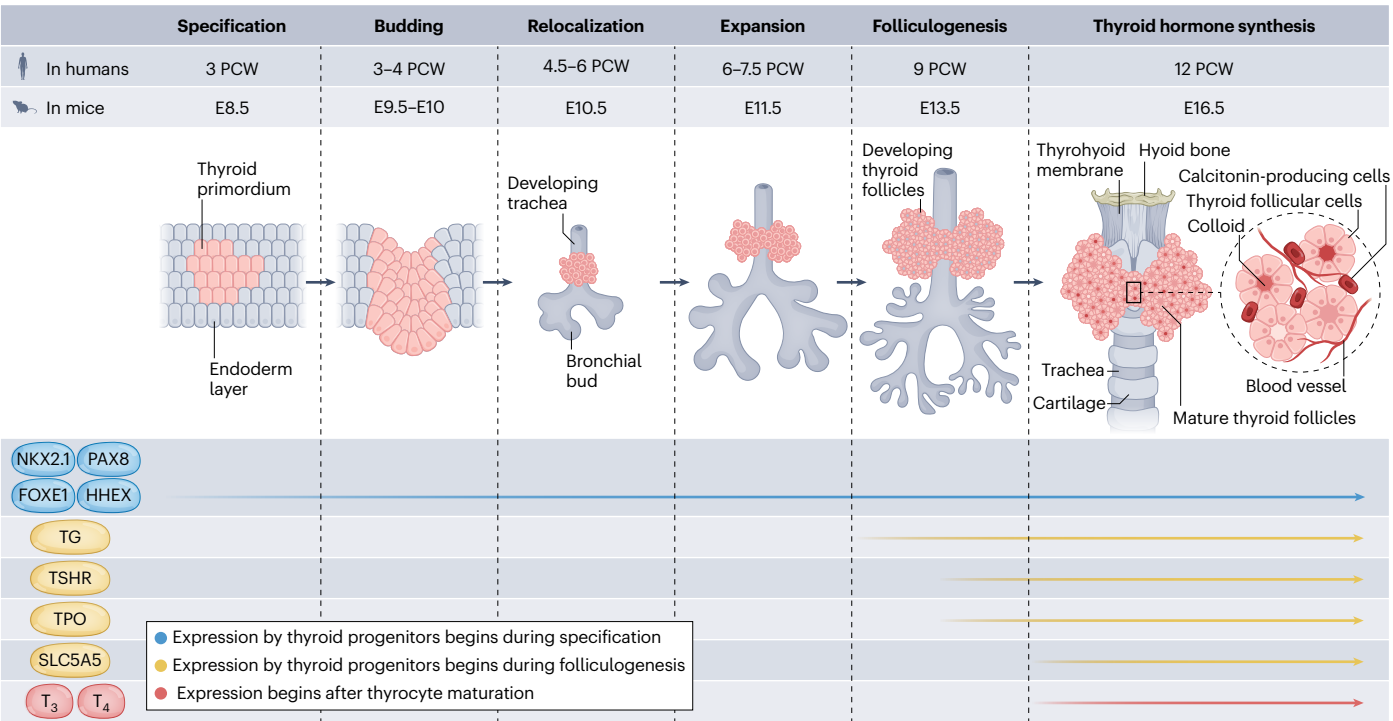


Fig. 1 | Schematic representation of thyroid developmental stages. The co-expression of NKX2.1, PAX8, FOXE1 and HHX characterizes the specification of thyroid progenitors from the endoderm layer. The thyroid primordium buds from the endoderm layer and relocates in front of the trachea. Subsequently, the thyroid progenitor population begins to expand by active proliferation. After expansion, thyrocytes organize into follicular structures, characterized by a monolayer of epithelial cells surrounding a luminal space, and express the earliest thyroid differentiation markers, namely, thyroid-stimulating hormone receptor (TSHR), thyroglobulin (TG) and thyroid peroxidase

(TPO), followed by the functional marker sodium–iodide cotransporter (SLC5A5). Following follicle formation, the thyroid tissue starts to produce tri-iodothyronine (T₃) and tetraiodothyronine (T₄), which characterize the final maturation and functional stage of the gland. The thyroid hormones are then released into the blood vessels surrounding the follicular structure. Together, follicles and vessels constitute the angio-follicular unit. In addition to the epithelial thyrocytes, the thyroid is also composed of calcitonin-producing cells (C cells). E, embryonic day; PCW, post-conception weeks.

in drug screening, and in toxicity studies. However, species differences pose considerable challenges for understanding human-specific processes. Conversely, human cell lines (2D) lack the structural complexity needed to replicate the organization and function of an organ. Ex vivo tissue cultures (explants) retain cellular complexity and structure but are limited by difficulties in obtaining healthy tissue, maintaining long-term in vitro cultures and preventing dedifferentiation⁸⁵.

Organoids have emerged as a promising alternative to animal models and 2D cell cultures. Organoids are composed of multiple cell types and form complex 3D structures in vitro mimicking the in vivo architecture and function of organs^{85,86}. Organoids can be generated from PS cells or directly from tissues. Importantly, tissue-derived organoids can arise even when adult stem cells have not been unequivocally identified in that tissue, suggesting that organoid culture conditions can activate latent progenitors, expand rare stem-like cells or confer stem-like properties on differentiated cells^{87,88}. PS cell-derived organoids replicate various stages of embryonic development and produce large cell quantities, whereas tissue-derived organoids are restricted to regenerative tissues, yielding simpler and less abundant structures⁸⁹.

PS cell-derived thyroid organoids

The stem cell-derived model that is most often used to study embryonic development and tissue formation relies on the differentiation of embryonic stem (ES) cells. ES cells are isolated from the inner mass of the blastocyst embryo and are highly proliferative, capable of self-renewing and can differentiate into cells from the three embryonic germ layers⁹⁰. However, transplantation of ES cell-derived organoids is limited by immune responses from the host. This limitation was addressed nearly two decades ago with the advent of induced PS (iPS) cells. The first researchers to produce these cells pioneered somatic cell reprogramming in cell types such as skin fibroblasts, blood cells and keratinocytes by transiently overexpressing pluripotency-related transcription factors *Oct4*, *Sox2*, *Klf4* and *Myc*^{91,92}. Remarkably, a study conducted in 2022 achieved chemical reprogramming of human somatic cells into chemically induced iPS cells that displayed key features of ES cells⁹³.

Thyroid organoids derived from ES cells and iPS cells have shown the ability to form three-dimensional follicles capable of synthesizing thyroid hormones. The directed differentiation approach uses controlled treatments with morphogens to induce or repress signalling, guiding PS cells towards a determined cell fate⁹⁴. Despite following the steps of development more physiologically, these strategies present several limitations: they are time-consuming procedures, inefficient, lead to low-purity cultures showing a substantial proportion of unwanted or non-target cell types, and have high variability⁹⁵.

Efficient generation of multiple cell types has been achieved by forced expression of single or multiple transcription factors in a highly controlled manner^{95,96}. Studies have shown cell reprogramming of murine and human PS cells by transient or constitutive forced expression of single or multiple transcription factors, resulting in cell types from the three embryonic germ layers^{94–96}. In 2021, a library of human transcription factors identified single or multiple players involved in the production of neurons, fibroblasts, oligodendrocytes and vascular endothelial-like cells from human PS cells⁹⁷. As for other tissues, in vivo embryonic developmental studies have provided valuable insights into the transcription factors, morphogens and signalling pathways involved in thyroid specification, forming the basis for thyroid organoid development. By overexpressing thyroid transcription factors and/or manipulating signalling pathways, studies have demonstrated

the differentiation of mouse and human PS cells into thyroid cells. The various approaches, strengths and limitations will be discussed in detail in the next two subsections.

Thyroid organoid generation by directed differentiation. The first insights into generating thyroid cells from mouse ES (mES) cells were provided about two decades ago⁹⁸. Embryoid bodies differentiated from mES cells and treated with TSH showed potential for differentiating into thyroid-like cells. However, morphological organization was not observed⁹⁸. Using a similar approach, TSH-treated cultures enriched for cells expressing GFP-tagged TSHR formed follicle-like structures expressing SLC5A5 and capable of iodine uptake. However, low levels of *Tpo* and absence of *Tg* were described⁹⁹. Using a new strategy, mES cell-derived embryoid bodies were differentiated into definitive endoderm and further matured into TG-expressing cells by incubation with TSH, insulin and IGF1 (ref. 100). Activin A induced endoderm differentiation and improved *Tshr* and *Slc5a5* expression, but maturation with TSH and/or IGF1 did not result in TG-expressing cells¹⁰¹.

Despite limited prior progress in generating TFCs from mES cells, major advancements in replicating early thyroid development were made in 2012 (ref. 33). A stepwise protocol was developed using a knock-in mES cell line expressing GFP-tagged NKX2.1. Initially, mES cells were differentiated into definitive endoderm using activin A, followed by AFE induction through TGF β and BMP inhibition. Subsequently, AFE cells were treated with a cocktail of thyroid-induction factors (BMP4 and FGF2) to promote specification. This procedure resulted in 24% GFP-tagged NKX2.1-positive cells, a mixed population of thyroid and lung cells. Further treatment with TSH, NaI and IGF1 led to partial maturation of thyroid progenitors to *Tg*-expressing and *Tshr*-expressing cells³³. To discriminate thyroid and lung lineages, a dual-fluorescent mouse iPS cell line expressing GFP-tagged NKX2.1 and Tomato-tagged PAX8 was used. Following definitive endoderm and AFE induction and sequential withdrawal of individual factors used for specification, BMP4 and FGF2 were found to be crucial for the generation of NKX2.1 and PAX8 co-expressing cells. Sorted thyroid progenitors were cultured in Matrigel using serum-free synthetic maturation media. At day 30, 3D-organized thyroid follicles showed TG and small amounts of T₄ in the lumen. In vivo transplantation of PS cell-derived thyroid cells into the kidney capsule of hypothyroid mice led to a substantial increase in levels of T₄ accompanied by TSH reduction after 8 weeks, maintained for at least 29 weeks³⁴.

Translating the mouse protocol to human cells, patient-derived iPS cells (from three patients carrying *NKX2.1* mutations) were differentiated using the strategy described in the preceding paragraph. NKX2.1-positive, PAX8-positive progenitors were derived after 10 days of treatment with BMP4 and FGF2. Thyroid progenitors were then cultured in maturation media, and TG, *TPO* and *TSHR* expression could be observed at day 42 (ref. 34). However, follicular organization and functionality were not shown. The critical role of BMP4 and FGF2 in human and mouse thyroid specification was confirmed and transcriptionally characterized in 2017¹⁰². However, a 2024 study using a double human iPS cell reporter line expressing GFP-tagged NKX2.1 and tdTomato-tagged PAX8 demonstrated highly efficient generation of thyroid progenitors by modulating the time of endoderm induction and the concentration of BMP4 treatment. The derived progenitors could be further matured into organized TFCs expressing TG, *TSH* and *TPO*. Nevertheless, T₄ production was not observed in vitro or after in vivo transplantation, which might be explained by the low levels of SLC5A5 expression³⁶.

TAZ is a known co-activator for *PAX8* and *NKX2-1* expression. Human ES (hES) cell-derived endodermal cells treated with TAZ activator (ethacridine) showed increased *NKX2.1* and *PAX8* expression. Further maturation with TSH enhanced levels of *TG*, *TPO*, *TSHR* and *SLC5A5*, and thyroid follicle-like structures capable of radioiodine uptake and organification were observed at day 21 (ref. 37). Using a similar protocol, the same team differentiated human dermal fibroblast-derived iPS cells into thyroid follicle-like structures with intraluminal TG. However, the ability to produce T_4 in vitro was extremely limited, and was essentially no better than that of FRTL5 rat thyroid cells¹⁰³.

Thyroid organoid generation by transcription factor overexpression. Thyroid progenitors are characterized by the unique co-expression of *NKX2.1*, *PAX8*, *FOXE1* and *HHEX*². Taking advantage of this knowledge, in a study conducted in 2012, our laboratory showed that the transient co-expression of *Nkx2.1* and *Pax8* (using a doxycycline-inducible system) in mES cells leads to thyroid specification⁵⁸. Subsequently, TSH treatment (using fetal bovine serum-containing differentiation media) led to the expansion and maturation of thyroid progenitors following the gene expression dynamics observed in vivo. At day 22, 3D-organized follicles showed iodine organification and intraluminal accumulation of the thyroid hormone precursor TG-I. Transplantation of the thyroid organoids into the kidney capsule of thyroid-ablated mice resulted in T_4 -producing follicles surrounded by blood vessels after 4 weeks. Serum levels of T_4 and TSH showed complete recovery of hypothyroidism, and symptoms also disappeared⁵⁸. Single-cell RNA sequencing analysis of the organoids revealed that, in addition to thyroid cells, they also contained non-thyroid cells, including fibroblasts, ES cells, and endothelial, epithelial, immune, neuronal-like and mesodermal cardiovascular cells¹⁰⁴. Interestingly, mesodermal cells were shown to affect thyroid function via TGF β activation¹⁰⁴.

Studies published in 2023 and 2024 combined microfluidic design and organoid technology to establish a thyroid organ-on-a-chip model^{105,106}. Mature mES cell-derived thyroid organoids cultured in an organ-on-a-chip microfluidics system showed a notable increase in the proportions of functional follicles compared with organoids grown in static conditions^{105,106}. Efficient thyroid progenitor derivation from mES cells was achieved by stage-specific transient overexpression of *Nkx2.1* followed by directed differentiation. By treating previously differentiated AFE cells with BMP4 and FGF2 and increasing the expression of *Nkx2.1* in a specific window of developmental time, *NKX2.1*-positive, *PAX8*-positive cell generation was achieved with an efficiency close to 60%. These progenitors were then 3D-cultured using a combination of factors³⁴, leading to T_4 -producing follicles³⁵. Other studies using stable *Nkx2.1* and *Pax8* overexpression in pre-induced mES cell-derived and hES cell-derived endoderm have demonstrated thyroid differentiation and iodine uptake. However, functionality was not demonstrated^{107,108}.

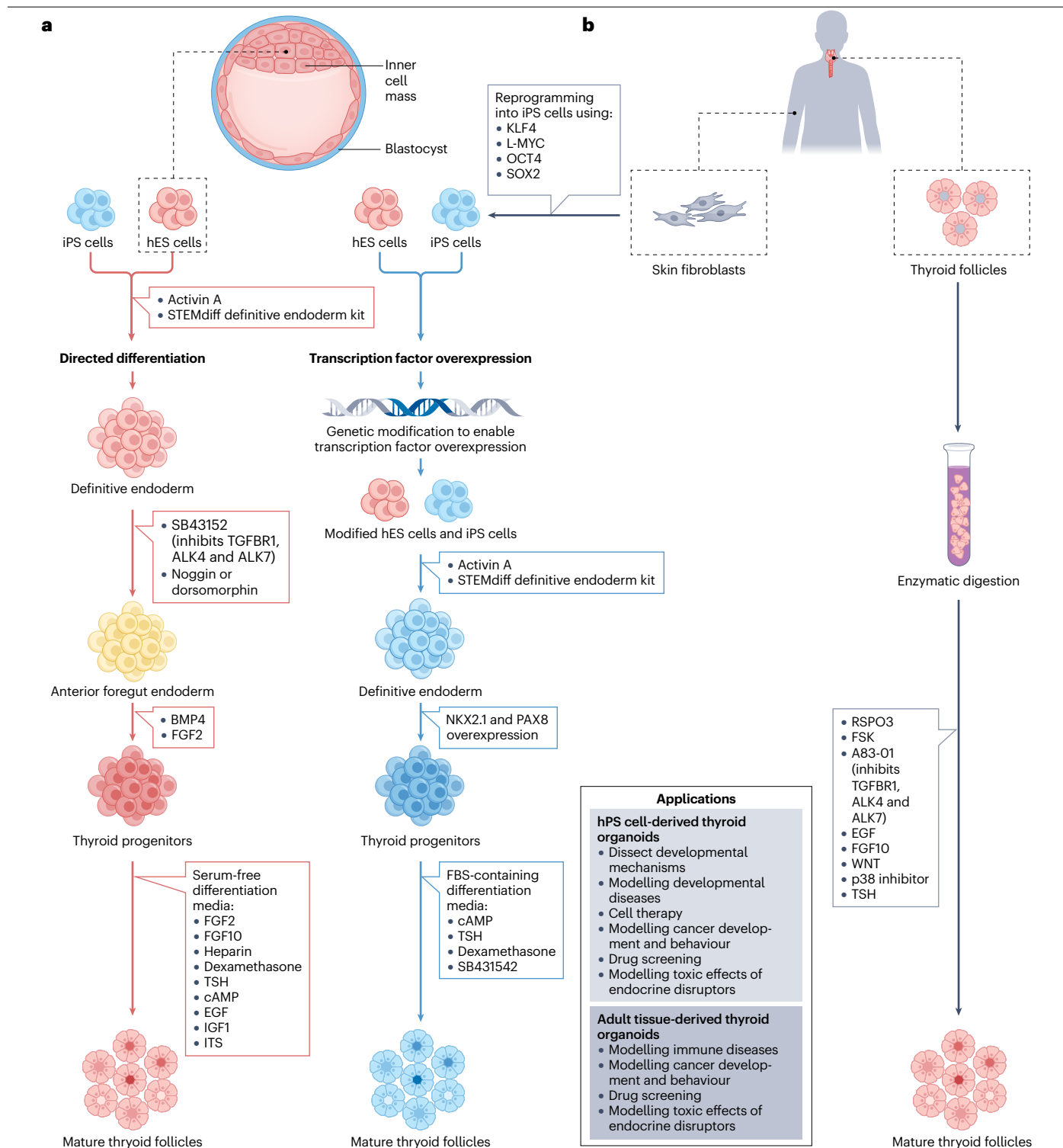
In 2022, our laboratory generated a protocol combining directed differentiation of hES cells into definitive endoderm and subsequent transient overexpression of *NKX2.1* and *PAX8*, efficiently deriving thyroid progenitors⁴⁸. Treatment with cAMP, in a fetal bovine serum-containing differentiation media, led to the expansion of the thyroid progenitor population, and recombinant human TSH, dexamethasone and TGF β inhibition promoted maturation. By day 58, 3D-organized follicles with intraluminal T_4 accumulation were observed. hES cell-derived thyroid follicles were then transplanted under the kidney capsule of NOD-SCID thyroid-ablated mice. After 5 weeks, T_4 -producing thyroid follicles surrounded by blood vessels were detected in the transplanted

site, and plasma levels of thyroid hormones and TSH were restored⁴⁸. Like the mES cell-derived organoids, the hES cell-derived organoids also contain fibroblasts and cardiovascular, epithelial, and airway cells. Interestingly, the thyroid population was found near cardiomyocytes, and cell–cell interaction predictions indicated potential crosstalk between thyroid and cardiovascular cells through various signalling pathways. This finding suggests that, like in vivo, during early in vitro development, thyroid progenitors might benefit from mesodermal cell signals, promoting proper specification and maturation.

Tissue-derived thyroid organoids

In contrast to the PS cell-derived organoids that model organ development, tissue-derived organoids recapitulate mainly tissue repair^{89,109}. Organoids can only be derived from tissue compartments with regenerative potential. Tissue-derived organoids are mainly restricted to the epithelial compartment of the organ, lacking tissue environment components like stroma, neural connections, immune cells and blood vessels^{86,89}. Tissue-derived organoids resemble the maturity observed in adult tissue but show smaller follicular structures, whereas PS cell-derived organoids are closer to a fetal state of maturation^{85,86,89}. The growing knowledge of tissue-specific signalling pathways was instrumental for adult stem cell maintenance and, alongside 3D culture systems, enabled the culture of adult stem cells and the generation of tissue-derived organoids for most epithelial organs^{85,86,109}. Healthy or diseased tissues (from biopsies, resections or bronchoalveolar lavage) are dissociated into single cells or small clumps, embedded in a 3D matrix, and cultured with tissue-specific factors essential for the stem cell niche^{85,86,109}. Within a few days, epithelial structures mimicking the architecture and function of the original epithelium form; these structures can be expanded in the long term while maintaining genetic stability⁸⁹.

The existence of thyroid adult stem cells remains under debate without a consensus^{22,110–114}. Nevertheless, organoids derived from adult and fetal thyroid tissues of rodents and humans have been successfully generated, recapitulating the gland's follicular architecture and function in vitro^{113,115,116}. Single GFP-positive and GFP-negative mixed thyroid cells from murine thyroid glands develop into exclusively GFP-positive or GFP-negative organoids, indicating that each organoid has single-cell origin¹¹⁵. Thyroid organoids expressing TG and pendrin and capable of thyroid hormone synthesis were generated and could be cultured and passaged for at least 6 months¹¹⁵. Dispersed murine and human thyroid cells cultured in Matrigel with chemically defined media (a culture medium where every component and its concentration are precisely known) formed thyrospheres expressing key thyroid markers and containing intraluminal T_4 . Following in vivo transplantation into thyroid-ablated mice, murine tissue-derived organoids formed functional follicles within 8 weeks, and human organoids required 26 weeks to achieve similar functionality¹¹⁶. A 2021 study demonstrated that 3D-cultured adult thyroid fragments generated organoids within 4 days for murine tissue and 1–2 weeks for human tissue. They self-organized into polarized follicles and secreted T_3 . Single-cell transcriptomics of mouse organoids revealed a small cluster of *Tg*-low proliferative cells, potentially representing elusive thyroid stem cells. However, no stem cell markers were identified, suggesting that thyroid tissue and organoids lack professional stem cells. Instead, organoid growth might rely on a transient proliferative subpopulation closely resembling mature TFCs¹¹³. Fetal thyroid organoids (FTOs) have also been derived from humans at 12 and 16 PCW¹¹⁷. FTOs also showed follicular organization and T_4 secretion¹¹⁷. Transcriptomic analysis



demonstrated that FTOs resemble the human thyroid differentiation and maturation steps *in vitro*. Transplantation of FTOs into the kidney capsule of mice showed numerous follicles able to secrete T_4 after 7 weeks¹¹⁷.

Compared with PS cell-derived organoids, tissue-derived thyroid organoids develop faster, require a simpler protocol and efficiently

generate functional follicles that can be expanded in the long term. Although they are limited in modelling developmental processes, they are highly effective for studying thyroid physiology, thyroid hormone production defects, immune diseases, carcinogenesis, drug responses and toxicity. However, the limited availability of healthy thyroid tissue remains a major constraint.

Fig. 2 | Schematic representation of PS cell-derived and tissue-derived thyroid organoid protocols.

a, Pluripotent stem (PS) cell-derived thyroid organoids: thyroid organoids can be derived from PS cells, which can be human embryonic stem (hES) cells or induced pluripotent stem (iPS) cells. iPS cells are typically reprogrammed from adult cells such as skin fibroblasts. Differentiation begins with the generation of definitive endoderm. In directed differentiation, which involves culturing the cells in media containing factors that promote specific differentiation pathways, this step is followed by the formation of anterior foregut endoderm, creating a niche conducive to thyroid progenitor specification. The transition from anterior foregut endoderm formation to thyroid progenitor specification is promoted by using chemical inducers such as FGF2 and BMP4. An alternative strategy involves using genetically modified PS cells in which the overexpression of transcription factors NKX2.1 and PAX8

can be induced following definitive endoderm formation. This approach leads to thyroid progenitor specification without the intermediate stage of anterior foregut endoderm formation. Subsequent maturation is achieved using specific maturation factors in either serum-free or serum-containing media, depending on the protocol. Complete maturation, including tetraiodothyronine (T_4) hormone production, has been reported in optimized protocols. **b**, Adult-tissue-derived thyroid organoids are generated by enzymatic dissociation of thyroid tissue into single cells or cell clusters. When cultured in a 3D matrix, these cells rapidly organize into follicles capable of producing thyroid hormones. Due to intrinsic differences, PS cell-derived and adult-tissue-derived organoids can be specifically used to tackle distinct questions about development and disease mechanisms as well as in cell therapy applications. cAMP, cyclic AMP; FBS, fetal bovine serum; TSH, thyroid-stimulating hormone.

The key features of each thyroid organoid model are summarized in Supplementary Table 1 and schematically shown in Fig. 2, with comparisons across approaches and studies also highlighted in Box 1.

Modelling thyroid diseases using organoids

Organoid technology offers a powerful tool to study cellular processes, with high potential in thyroid translational research. Here, we highlight its utility in modelling thyroid diseases.

Hypothyroidism

Hypothyroidism affects 1–5% of the population worldwide. It is a consequence of insufficient thyroid hormone synthesis caused by genetic factors or developmental disorders, autoimmune diseases, iodide excess or deficiency, and radiation. Hypothyroidism can also result from total thyroidectomy performed for diseases such as Graves' disease or goitre, malignant conditions like thyroid cancer, or, more rarely, from central hypothyroidism, a disorder caused by impaired thyroid hormone production due to deficient TSH secretion from the pituitary or thyrotropin-releasing hormone secretion from the hypothalamus^{118–121}. In the USA, roughly 130,000–169,000 thyroid surgeries are performed annually¹²². In Europe, about 45,000 such procedures occur each year in France, 60,000 in Germany and 4,000 in Switzerland¹²³. Globally, of the 586,000 people diagnosed with thyroid cancer in 2020¹²⁴, around 90% are predicted to undergo total thyroidectomy¹²⁵. Consequently, thyroidectomy is a major contributor to the population of patients with hypothyroidism.

Congenital hypothyroidism, the most frequent neonatal metabolic disorder (1:1,500–4,000 live births), has a severe effect on neural and skeletal development if untreated^{126–129}. Most congenital hypothyroidism cases (85%) are due to impaired thyroid organogenesis and migration, which lead to thyroid dysgenesis, and the remaining cases (15%) are associated with an inability to produce normal amounts of thyroid hormones (dyshormonogenesis) with gland in situ^{15,127,130–133}. Less than 10% of thyroid dysgenesis cases are associated with mutations in genes implicated in thyroid morphogenesis. The genetics of congenital hypothyroidism has been extensively reviewed^{127,132–134}. Next-generation sequencing data have revealed one or multiple genetic variations in more than 50% of patients with congenital hypothyroidism and dyshormonogenesis¹³⁴. However, the effect of those variants on thyroid development and function is unclear, and organoids emerge as a powerful tool to answer these open questions.

iPS cells from children with hypothyroidism who had brain–lung–thyroid syndrome and *NKX2-1* mutations showed reduced efficiency in generating thyroid progenitors, confirming the detrimental effect of

Nkx2-1 haploinsufficiency observed in mouse models^{17,18,34,42}. Mutations in *FOXE1* are associated with severe hypoplasia or complete agenesis of the thyroid gland^{142,135}. *Foxe1*^{−/−} mES cells fail to efficiently differentiate into thyroid progenitors, resulting in few non-functional thyroid follicles¹³⁶. *Nrf2*^{−/−} (leading to knockout of nuclear factor erythroid 2-related factor 2 (NRF2)) mES cells properly differentiate into thyroid progenitors, although they showed impaired *Tg* expression and thyroid hormone production¹³⁷. NRF2 is a master regulator of the cellular oxidative and metabolic stress defence system^{138,139}, and is suggested to have an important role in helping the thyroid to resist H_2O_2 -induced oxidative stress while maintaining the homeostasis necessary for hormone synthesis.

Effect of toxic compounds: focus on endocrine disruptors

Endocrine disruptors are chemicals that display an 'endocrine mode of action' that can disrupt endocrine system function^{140,141}. Increasing evidence suggests that endocrine disruptors have serious effects on the thyroid axis, altering circulating levels of thyroid hormones, disrupting thyroid hormone metabolism and signalling in target tissues, and inducing thyroid cytotoxicity¹⁴⁰. Nevertheless, the mechanisms of action of potential endocrine disruptors across different endocrine tissues remain poorly understood. Two-dimensional cultures of primary thyroid cells and immortalized cell lines are the *in vitro* tests approved by the Organisation for Economic Co-operation and Development¹⁴² for toxicity studies of endocrine disruptors. However, maturation genes are downregulated in most of these lines, which in turn affects their relevance for toxicity studies.

A pioneering study conducted in 2015 tested the effect of benzo[k]fluoranthene (a polycyclic aromatic hydrocarbon) exposure on mES cell-derived thyroid organoids¹⁰⁶. Using a microfluidics organ-on-a-chip culture device, thyroid organoids exposed to benzo[k]fluoranthene for 24 h showed downregulation of *Tpo*, *Slc5a5* and *Duoxa2* genes, suggesting an effect on thyroid maturation. In another study, thyroid organoids acutely exposed to low doses of the phthalates bis(2-ethylhexyl)phthalate, di-iso-decylphthalate, di-iso-nonylphthalate and di-noctylphthalate for 24 h did not show changes in the expression of the main thyroid genes¹⁴³. These results support the hypothesis that the harmful effects of endocrine disruptors in people might be due to long-term exposure and cumulative effects. In a later study¹⁴⁴, the role of male versus female sex hormones in thyroid responses to two endocrine disruptors, benzo[a]pyrene and PCB153, was assessed in human thyroid organoids. Compared with organoids that were exposed to benzo[a]pyrene, but not to sex hormones, organoids that were exposed to both benzo[a]pyrene and either male or female sex

Box 1 | Approaches for generating thyroid organoids from PS cells

Two main strategies are used to generate thyroid organoids from pluripotent stem (PS) cells: directed differentiation via stepwise chemical cues and transcription factor overexpression. This box summarizes the main protocols able to produce mature organoids.

Directed differentiation

- Starting cells: human or mouse PS cells (embryonic stem (ES) cells or induced PS cells)^{33,34,36}.
- Approach: mostly 3D culture based on sequential exposure to defined signalling molecules that mimic thyroid development.
- Timeline: from 3–4 weeks (mouse PS cells)³³ to 6–7 weeks (human PS cells)^{34,36}.
- Media: serum-free, chemically defined, stage-specific media with growth factors and small molecules.
- Outcome: follicular thyroid-like cells that self-organize into 3D follicles, express thyroid-maturation genes and can produce thyroid hormones in vitro and in vivo^{33,34}.
- Advantages: mimics natural developmental cues. No genomic manipulation.

- Limitations: efficiency can be variable, and functionality has not been obtained from human PS cells³⁶.

Transcription factor overexpression^{58,114}

- Starting cells: genetically modified human or mouse ES cells.
- Approach: 3D culture based on forced transient expression of key thyroid transcription factors, such as NKX2.1 and PAX8, in pre-induced endodermal cells.
- Timeline: from 3 weeks (mouse ES cells) to 8–10 weeks (human ES cells).
- Media: serum-containing media supplemented with minimal supporting factors.
- Outcome: cells acquire thyroid-specific gene expression, express functional markers and form follicles capable of producing thyroid hormones in vitro and in vivo.
- Advantages: high efficiency of lineage specification and organoid generation.
- Limitations: requires genomic manipulation, which results in bypassing the specification stage.

hormones showed differential gene expression. Male hormones and female hormones each elicited distinct transcriptional responses in benzo[a]pyrene-treated organoids. Compared with organoids that were treated with PCB153 alone, organoids that were also exposed to male hormones showed a modest difference in gene expression but organoids exposed to female hormones and PCB153 had a similar transcriptional profile to organoids that were only exposed to PCB153 (ref. 144). These results suggest that hormonal status and sex could add a layer of complexity to the response to endocrine disruptors. Ideally, both male-derived and female-derived organoids should be used to explore the role of genetic background in responses to endocrine disruptors.

Thyroid cancer and patient-derived tumouroids

The 3D organoid system has emerged as a valuable alternative to bridge the gap between conventional 2D cell lines and animal models in cancer research. Differentiated thyroid cancers originating from follicular cells are classified into two subtypes: papillary thyroid cancer (PTC), the most common form (accounting for ~80% of cases), and follicular thyroid cancer. Poorly differentiated thyroid carcinoma (PDTC) and anaplastic thyroid carcinoma (ATC) represent the most advanced and aggressive forms, with ATC also referred to as the undifferentiated type^{145,146}. Although the vast majority of differentiated thyroid cancers respond favourably to standard treatments, primarily surgery followed by radioactive iodine (RAI) therapy, a considerable subset (~20% of PTCs) either partially or completely loses the ability to take in iodine. These RAI-refractory (RAIR) tumours exhibit reduced responsiveness to conventional therapies and are associated with a considerably poorer prognosis. In contrast to differentiated thyroid cancers, PDTC and ATC display markedly aggressive clinical behaviour from the outset. These tumours are characterized by rapid progression, early metastasis, and intrinsic resistance to RAI and most systemic treatments, making them exceptionally difficult to manage. Despite representing a minority of thyroid cancer diagnoses (2–5% of thyroid cancer cases), PDTC and especially ATC account for up to 50% of thyroid cancer-related deaths,

underscoring their disproportionate clinical impact. Medullary thyroid carcinoma (MTC), by contrast, arises from parafollicular C cells and is a rare subtype (<5% of thyroid cancers). As for follicular-derived thyroid cancer, resection remains the standard of care for patients with MTC; however, recurrence has been reported in ~50% of patients and is associated with worse outcomes¹⁴⁷.

Advances within the past decade in thyroid cancer management, complementing standard therapies with chemotherapy, immunotherapies and/or molecularly targeted agents, have substantially improved outcomes for subsets of patients by enabling greater diagnostic precision and more individualized treatment strategies based on tumour genetics and subtype^{148,149}. In particular, PDTC and ATC remain major clinical challenges due to their aggressive behaviour and limited responsiveness to current therapies. This challenge underscores the urgent need to elucidate the molecular and cellular mechanisms driving PDTC and ATC development and progression, an essential consideration for the rational design of more effective therapeutic strategies.

To address these gaps and explore the carcinogenic process from early cell transformation to drug response, patient-derived tumouroids, also known as cancer spheroids, have been established, providing powerful tools for target discovery and preclinical testing¹⁵⁰. Members of the cancer research community have now proposed that a more relevant term for 3D models derived from cancer tissues is ‘tumouroids’, which encompasses both cancer organoids and spheroids. The term organoid traditionally refers to miniature, organ-like structures that recapitulate the architecture and physiological function of healthy tissues, whereas cancer-derived models rarely retain such organized structures or specialized functions. Given that tumours are defined by uncontrolled proliferation, intra-tumour cellular heterogeneity and structural disorganization, the term tumouroids (especially when compared to thyroid organoids, which contain a unique cell type and are structurally well organized into follicles) more accurately reflects their pathological origin while maintaining conceptual continuity with existing organoid and spheroid methodologies^{151–153}. In the thyroid cancer field, this terminology has not yet been widely adopted;

most studies continue to refer to thyroid cancer organoids or thyroid cancer spheroids when describing 3D cultures derived from malignant tissue. Here, we propose adopting the term ‘thyroid tumouroids’ to better convey the pathological nature of these models, clarify their distinction from healthy thyroid organoids and align thyroid cancer research with emerging standards in other cancer organoid fields.

Thyroid tumouroids derived from PS cells. To model thyroid cancer development, studies have used ‘normal’ thyroid organoids to introduce common human oncogenic mutations. Expression of *NRAS*^{Q61R} in thyroid organoids derived from *Tp53*^{-/-} mice generated PDTc after transplantation¹¹⁵. Similarly, induction of *Braf*^{V637E} (murine equivalent of human *BRAF*^{V600E}) expression in mES cell-derived thyroid organoids produced a PTC-like phenotype in vitro, which could be reversed by combined MAPK and PI3K inhibition, restoring follicular organization and function¹⁵⁴. Using hES cell-derived thyroid progenitors and mature thyrocytes, the oncogenic potential of *BRAF*^{V600E} or *NRAS*^{Q61R}, alone or with *TP53*^{R248Q}, was evaluated¹⁵⁵. Surprisingly, *BRAF*^{V600E} or *NRAS*^{Q61R} expression in mature thyrocytes showed limited carcinogenic capacity. However, when induced in thyroid progenitors, PTC and follicular thyroid cancer phenotypes were observed. *TP53*^{R248Q} in association with *BRAF*^{V600E} or *NRAS*^{Q61R} in thyroid progenitors generated undifferentiated thyroid cancers¹⁵⁵. More recently, human thyroid organoids with germline *PTEN* mutations were used to investigate early carcinogenic responses¹⁵⁶. They uncovered a divergent *PTEN*–*TP53* interaction upon DNA damage, showing that *PTEN*-mutant organoids activate aberrant p53 signalling. These results illustrate how genetic background reshapes early stress-response pathways and reinforce the value of PS cell-derived models for uncovering genotype-specific vulnerabilities and previously unrecognized tumour-suppressor interactions.

PS cell-derived thyroid tumouroids have advanced our understanding of early thyroid carcinogenesis. By enabling precise introduction of oncogenic mutations into defined cell states, these systems show that the same lesions can produce distinct outcomes in progenitors versus mature thyrocytes. Although PS cell-derived models suggest that thyroid progenitors are particularly susceptible to transformation, this view remains controversial, as in vivo evidence indicates that tumours originate from mature thyroid cells. These discrepancies highlight the need for further studies, including lineage-tracing and in vivo validation, to clarify the true cells of origin in thyroid cancer. These models also uncover mutation-specific alterations in stress-response and tumour-suppressor pathways, providing a unique window into the molecular events that precede overt transformation. PS cell-derived thyroid tumouroids allow the identification of early dysregulated signalling networks and candidate drivers that could serve as targets for therapeutic intervention. Future applications integrating CRISPR-based perturbations, single-cell transcriptomics and epigenomics, and lineage-tracing strategies will enable high-resolution mapping of the molecular rewiring that drives thyroid cells towards malignancy, refining our understanding of tumour initiation and pinpointing actionable targets before tumours become genetically complex and clinically intractable.

Patient-derived thyroid tumouroids. Tumouroids can be efficiently derived from patient tumour samples, recapitulating the histological and molecular characteristics of the original tumour. As a result, patient-derived thyroid tumouroids have become a valuable preclinical model to investigate tumour heterogeneity, progression, therapy response and drug resistance¹⁵⁷.

PTC organoids have been generated with >70% efficiency from primary tumours^{158–163}. Primary tumour samples and derived tumouroids show conserved histological characteristics and genome stability even at later passages¹⁵⁸. Preclinical studies using multiple anticancer drugs demonstrated patient-specific drug responses and a high level of correlation with their mutational profiles¹⁶³. Interestingly, tumouroids derived from distinct areas of the same primary tumour show different drug responses, suggesting intratumoural heterogeneity¹⁶³. *BRAF*^{V600E}-mutated tumouroids also demonstrated improved drug response when combining *BRAF* inhibitors with MEK inhibitors or tyrosine kinase inhibitors or with chemotherapy¹⁵⁸. Because these tumours are largely resistant to conventional therapies, tumouroids derived from RAIR PTC, PDTc and ATC are of particular interest, providing essential platforms to investigate therapeutic responses and uncover new molecular targets. RAIR PTC tumouroids (characterized by the lack of *SLC5A5* expression) were shown to be substantially discordant at protein and gene expression levels when compared with PTC organoids¹⁶⁴, underscoring the value of these models for screening novel therapeutic strategies.

Thyroid tumouroids derived from fine-needle aspiration (FNA) samples have also been described¹⁶². FNA is a minimally invasive technique widely used as a first-line procedure for sampling and diagnosing thyroid tumours. FNA sample-derived PTC and ATC organoids preserved the primary tumour histology while maintaining a tumour microenvironment compartment composed of stromal and immune cells¹⁶². Using this method, two patient-derived ATC tumouroid models (*BRAF*^{WT} and *BRAF*^{V600E}) were established in 2024 and used to evaluate the effect of a WNT– β -catenin inhibitor, pyrvinium pamoate. Interestingly, the researchers observed considerably reduced tumouroid growth both with pyrvinium pamoate alone and in combination with standard therapies¹⁶⁵. This result further highlights the use of tumouroids derived from FNA samples to inform personalized therapy, allowing drug response assessments before and after treatment. However, considering the expected tumour cell heterogeneity, multiple FNA-derived tumouroids should be used to predict tumour response and define therapeutic strategies.

In 2025, a landmark study demonstrated the translational power of thyroid tumouroids. In this prospective phase II study, tumouroids derived from locally advanced thyroid cancer were used in drug sensitivity tests that directly guided clinical decision-making¹⁶⁶. Biopsy samples of primary or metastatic lesions from 75 patients were used to generate tumouroids and assess individualized therapeutic responses in vitro; 55 patients received targeted therapies that were selected based on tumouroid-derived sensitivity profiles and achieved an objective response rate of 32.7% (differentiated thyroid cancers 32.6%, MTC 16.7% and ATC 50%). Of these patients, 24 proceeded to surgical resection and 19 underwent complete tumour removal. This study underscores the potential of tumouroid-guided treatment choices to improve overall outcomes. It also highlights the variability among cancer subtypes, suggesting that despite improvements in treatment response predictability, the current tumouroid models can still fail to fully recapitulate the complexity of the original tumour. Potential limitations include culture-driven selection of specific cell populations, loss of intratumoural heterogeneity, suboptimal growth conditions and the absence of key microenvironmental components such as vascular, stromal and immune cells, which critically shape tumorigenesis and therapy response. Addressing these gaps through improved culture systems and integration of microenvironmental elements will be essential for building more physiologically relevant thyroid cancer models

Box 2 | Challenges in transferring thyroid organoids from the bench to the bedside

- Standardization of organoid generation; this can be achieved by optimizing and implementing standardized analysis methods and quality control assessments.
- Genetic and epigenetic variability and the need to develop safe methods of reprogramming induced pluripotent stem cells.
- Thyroid organoid transplantation requires a highly vascularized, easily accessible site where surgery poses low levels of risk to the patient and the graft can be easily monitored.
- Successful organoid engraftment and function depend on a number of factors such as the patient's age and any pre-existing health conditions.
- The immune response to transplanted thyroid organoids also poses a challenge and can necessitate chronic immunosuppression. Allogeneic transplantation, the use of hypoinmunogenic organoids and using microencapsulation to reduce organoid immunogenicity are other approaches to surmounting this challenge.

Translating thyroid organoid production into a clinical setting will require highly efficient directed differentiation protocols. Identifying key differentiation factors, boosting thyroid progenitor generation and achieving full thyroid maturation will be important aims during the development of these protocols.

capable of accurately predicting patient responses and guiding clinical decision-making.

A 2025 study established patient-derived medullary thyroid tumouroids that preserve the tumour's gene and protein signature, including secretion of calcitonin and CEA. The study further demonstrated that these tumouroids can be used to evaluate both tyrosine kinase inhibitor effects and positron emission tomography and single-photon emission computed tomography imaging tracer uptake, illustrating their potential for in vitro therapy testing and imaging tracer evaluation¹⁶⁷. In a broader thyroid cancer context, this approach could be applied to test RAI responsiveness in differentiated tumours or to develop and optimize new tracers, such as radiolabelled peptides targeting RET in MTCs. By enabling ex vivo evaluation of tracer behaviour in patient-derived tumouroids, these systems provide a translational platform for personalized imaging strategies and therapy planning, bridging molecular diagnostics with individualized patient care.

Cell therapy: how close are we?

Curative approaches are preferable to lifelong hormone replacement, and in hypothyroidism, the restoration of a fully functional thyroid gland could re-establish physiological, demand-driven thyroid hormone synthesis and release. Although thyroid transplantation represents a promising curative strategy for every patient, its application must be considered carefully, as most patients currently achieve satisfactory outcomes with a low-risk, low-cost standard therapy. Nevertheless, thyroid tissue replacement could be particularly valuable for patients who respond inadequately to conventional treatment. Notably, about one-third of patients with hypothyroidism do not receive adequate treatment, and a large proportion experience impaired psychological well-being^{168–171}. Social disorders, along with

motor and cognitive impairments, have been reported in children with congenital hypothyroidism, even with early diagnosis and thyroid hormone replacement therapy¹⁶⁹. During puberty, growth, stress and pregnancy, fluctuations in endogenous levels of thyroid hormone occur, which cannot be fully replicated by exogenous stable doses^{120,172}. In general, thyroid organoid transplantation might offer a more effective therapeutic approach, though clinical application remains a challenge. In the following subsections, we discuss the key obstacles to translating organoid models from the bench to the bedside (Box 2).

Enhancing development of functional thyroid organoids through directed differentiation

The approach that is currently the most efficient for generating functional human thyroid organoids relies on the overexpression of transcription factors and, consequently, on genomic manipulation⁴⁸. This limitation was partially solved by using a directed differentiation approach using chemical compounds and PS cells. Although this strategy generates high numbers of TFCs, functionality has not yet been achieved³⁶. Identifying additional key factors involved in thyroid maturation and function is crucial to producing a sufficiently 'rich' culture medium capable of generating fully functional thyroid organoids. Within the past 5 years, advances in single-cell transcriptomics^{36,48,173} enabled the exploration of gene expression signatures in thyroid progenitors, differentiating thyrocytes and mature thyrocytes, revealing key players at each stage of development and potentially contributing to the adaptation of in vitro protocols.

Host immune responses to organoid transplants

Transplant rejection driven by the immune system is a major obstacle to bringing organoid therapies to the clinic. In an allogeneic transplantation context, human leukocyte antigen (HLA) class I molecules can activate T cells of the recipient's immune system by presenting 'unknown' peptides from donor tissue, triggering a strong immune response and transplant rejection^{174,175}. Chronic immunosuppression prevents allogeneic transplantation rejection but increases risk of complications, infections and malignancies¹⁷⁶. To address these challenges, alternative strategies have been explored to prevent immune recognition and rejection. Mouse and human iPS cells engineered to lack major histocompatibility complex (MHC) class I and II genes while overexpressing *CD47* demonstrated reduced immunogenicity while maintaining pluripotency and differentiation potential. Hypoimmunogenic iPS cell-derived cardiomyocytes, endothelial cells and smooth muscle cells have successfully evaded immune rejection in fully MHC-mismatched allogeneic recipients without requiring immunosuppression¹⁷⁷. Despite the protection against T cells in vitro, HLA class I-knockout kidney organoids transplanted into humanized mice were rejected similarly to wild-type organoids, possibly due to compensation by HLA class II molecules¹⁷⁸.

iPS cell-derived human islet-like organoids overexpressing the immune-checkpoint protein programmed death-ligand 1 (PDL1) escaped immune response when transplanted into immune-competent diabetic mice while retaining the ability to restore glucose homeostasis¹⁷⁹. This observation highlights the need for a better understanding of the specific immune response to each organoid system and paves the path to a more reliable immune evasive approach¹⁷⁶. Despite promising results, these strategies rely on genetic modifications to prevent immune responses post-transplantation. Advances in bioengineering offer an alternative through encapsulation methods, involving the use of biocompatible polymers to create a selective barrier that isolates

organoids from immune cells, cytokines and antibodies while allowing oxygen, nutrient and hormone exchanges¹⁸⁰.

On the other hand, autologous iPS cells comprise an unlimited cell source for patient cell-based organ repair strategies. Non-integrating reprogramming methods offer a safer approach to generating iPS cells than methods that involve vector integration into the genome, enabling the development of patient-derived organoids for cell-based therapies¹⁸¹. However, iPS cell variations raise special concern due to differences between donor individuals, genetic stability, epigenetics and experimental variability, which affect differentiation yield, cellular heterogeneity, morphology, and gene and protein expression¹⁸². Beyond the inter-individual variations, patient-specific iPS cell-derived clones can be highly variable in pluripotency, affecting cell identity and organoid development^{182,183}. Implementing rigorous iPS cell reprogramming standardization, clone selection and characterization, along with precise differentiation protocols, can minimize inter-iPS cell variation and ensure reproducible organoid generation. As discussed in the section 'Thyroid organoid generation by directed differentiation', human thyroid organoids have been generated from iPS cells using directed differentiation methods^{34,36,103}. Yet, variations in the protocols and culture system result in highly variable efficiency of thyroid cell generation and maturation. Together, thyroid organoids derived from iPS cells through directed differentiation, combined with standardized protocols, can bring us closer to autologous transplantation of functional human thyroid organoids.

Transplantation site selection and inter-individual variations

Unlike most organoids, endocrine organoids, including the thyroid, can achieve full functionality without needing to be engrafted into their original anatomical site. The main requirement to release the thyroid hormones into the blood flow is the presence of blood vessels. Thyroid organoids were successfully transplanted into the kidney capsule of mice, showing surrounding vascularization while remaining stable and functional for several weeks^{34,48,58,116}. However, comprehensive safety studies are necessary to assess cell state, graft stability, functionality and long-term preservation. Translating this technique to humans poses risks, particularly due to the invasive surgical procedures required for kidney capsule transplantation. Identifying alternative, highly vascularized and easily accessible transplantation sites is crucial in reducing the risks and facilitating monitoring and potential graft removal. Additionally, the dynamic nature of thyroid hormone requirements across life stages constitutes a challenge in determining the optimal number of follicles needed to maintain target levels of thyroid hormone¹⁸⁴. Future human clinical trials must consider variables such as sex, age, weight, height and pre-existing conditions to ensure precise and personalized outcomes.

Other translational barriers

Bringing thyroid organoid models from the bench to the clinic faces several critical translational challenges. Ensuring the reproducibility, safety and scalability of iPS cell-derived products is essential. Compliance with good manufacturing practices and the development of robust, low-risk differentiation protocols are key prerequisites for clinical translation. Currently, most organoid cultures rely on animal-derived matrices (for example, Matrigel), which carry potential risks for patients, including immunogenicity or pathogen transmission. Alternatively, non-animal-derived biocompatible scaffolds, such as hydrogels, agarose, cellulose, or synthetic extracellular matrix analogues, offer safer options while maintaining the organoid's structural and functional

integrity. However, for thyroid organoids, such optimized systems have not yet been fully established, highlighting a major gap before clinical application. Additionally, tumorigenic risk remains incompletely characterized. Most studies to date lack long-term in vivo follow-up, making it difficult to definitively exclude the possibility of undesired tumour formation following transplantation of thyroid organoids. Addressing these issues will be essential to reaching the therapeutic and diagnostic potential of thyroid organoids in regenerative medicine and precision oncology.

Conclusions

Despite the challenges, human thyroid organoids hold immense promise for understanding development, modelling diseases and potentially serving as a therapeutic tool. Integrating thyroid organoids with advanced technologies, such as single-cell omics, CRISPR–Cas9 gene editing, co-culture systems and microfluidic organ-on-a-chip platforms, could enable comprehensive modelling of thyroid tissue, evaluation of disease mechanisms and identification of novel therapies. To translate these advancements into clinical applications, it is essential to unify efforts across research laboratories, implement rigorous standardization and adhere to good manufacturing practices to ensure safe and effective delivery from the Petri dish into a patient's body.

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Author contributions

The authors contributed equally to all aspects of the article.

Competing interests

S.C. and M.R. have filed an international patent regarding the human embryonic stem cell-derived thyroid organoids.

Additional information

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